

HINTS AND SOLUTIONS

Q.1 A

Sol. If $\sqrt{V}$ for k_α lines are plotted against atomic number it will be straight line

$$\sqrt{V} - \sqrt{V_1} = \left(\frac{\sqrt{V_2} - \sqrt{V_1}}{Z_2 - Z_1} \right) (Z - Z_1)$$

$$\text{or, } \left[\left(\frac{\sqrt{V_2} - \sqrt{V_1}}{Z_2 - Z_1} \right) (Z_3 - Z_1) + \sqrt{V_1} \right]^2$$

Q.2 A

Q.3 A

Sol. $\lambda_{1(\max)} = \frac{hc}{\Delta E_{2 \rightarrow 3}}$

$$\lambda_{2(\max)} = \frac{hc}{\Delta E_{3 \rightarrow 2}}$$

$$\lambda_{1(\max)} / \lambda_{2(\max)} = \frac{hc}{hc} \times \frac{\Delta E_{3 \rightarrow 2}}{\Delta E_{2 \rightarrow 3}} = 1$$

Q.8 AC

Sol. $\frac{P^2}{2m_\alpha} + \frac{P^2}{2m_{th}} = Q \Rightarrow P = \left(\frac{2m_\alpha m_{th} Q}{m_\alpha + m_{th}} \right)^{1/2}$ and $\frac{K.E_\alpha}{K.E_{th}} = \frac{m_{th}}{m_\alpha} = \frac{234}{4}$

Q.10 BCD

Sol._{vj} Intensity increases no. of photon coming on plates therefore it wouldn't change stopping potential work function maximum K.E.

Q.13 B

Sol. $\frac{1}{2} \mu v_r^2 \geq \Delta E$

$$\frac{1}{2} \frac{m}{2} (v - 0)^2 \geq 0.5$$

$$\therefore \frac{1}{2} mv^2 \geq 1 \text{ MeV}$$

Q.14 A

Sol. $Q = (m_T + m_H - 2m_p)C^2$
 $4 = (m_T + 1.0078 - 2 \times 2.0141) \times 931.5$
 $4.294 \times 10^{-3} = m_T + 1.0078 - 4.0282$
 $4.0282 - 1.0078 - 0.0043 = m_T = 3.0161 \text{ amu}$

Q.15 Ans: 100.26 days.

Q.16 Ans. $(1:2^{1/3})$

Sol. $R \propto A^{1/3}$

Where R radius of nuclei.

Q.17 Ans. $[(2)^{-5}]$

Sol. $\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/T} = \left(\frac{1}{2}\right)^{5T/T} = 2^{-5}$

Q.18 Ans. 227.61883amu

Sol. $(m_p - m_\alpha - m_D) \times 931.47 = (K_\alpha + K_D) \text{ in MeV} \quad \dots(i)$

$P_\alpha = h/\lambda_\alpha = 1.15 \times 10^{-19} \text{ kg-m/s}$

Since initial linear momentum of the system is zero, $P_\alpha = P_D$

$\therefore K_\alpha = P_\alpha^2 / (2m_\alpha) = 1 \times 10^{-12} \text{ J}; K_D = P_D^2 / (2m_D) = 1.78 \times 10^{-14} \text{ J}$

$\therefore \text{Total KE} = K_\alpha + K_D = 1.0178 \times 10^{-12} \text{ J} = 6.36 \text{ MeV}$

Putting it in equation (1)

$m_p = m_\alpha + m_D + \{(K_\alpha + K_D) \text{ in MeV}\} / 931.47 = 227.61883 \text{ amu}$

(Taking $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$)

Q.19 Ans. 4×10^{-6}

Sol. Velocity of neutrons $= \sqrt{\frac{2eV}{m_n}} = \sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 0.0327}{1.67 \times 10^{-27}}}$

$= \sqrt{6.29 \times 10^6} \text{ m/s} = 2.5 \times 10^3 \text{ m/s}$

Time taken by neutron to cover the distance 10 m

$t = \frac{10}{2.5 \times 10^3} = 4 \times 10^{-3} \text{ seconds} \ll T_{1/2}$

$\lambda = \frac{0.693}{700} = 9.9 \times 10^{-4} / \text{sec}$

Fraction decayed $= 1 - e^{-\lambda t} \approx \lambda t = 4 \times 10^{-6}$

Q.20 Ans. $Z = 42, 28$

Sol. According to Moseley's equation for k_α radiation ;

$\frac{1}{\lambda} = R(z-1)^2 \left(\frac{1}{1^2} - \frac{1}{2^2} \right)$

$\frac{\lambda_1}{\lambda} = \frac{(z-1)^2}{(z_1-1)^2} = \frac{0.7092}{1.5405}$

$z = 29 \text{ for cu, hence } z_1 - 1 = 28 \sqrt{\frac{1.5405}{0.7092}} = 41$

or $z_1 = 42$

$\therefore$ Impurity is molybdenum

similarly; $\frac{\lambda_2}{\lambda_1} = \frac{(z-1)^2}{(z_2-1)^2} = \frac{1.6578}{1.5405}$

or $z_2 - 1 = 28 \sqrt{\frac{1.5405}{1.6578}} = 27$

$z_2 = 28$

It is atomic number of Nickel.

Hence the other impurity is Nickel.

